Dr. A. P. J. Abdul Kalam Technical University, Lucknow



Evaluation Scheme & Syllabus

for

Bachelor of Design (B.Des.)

I, II, III & IV Year

(For Batch 2020 Onwards)

PREFACE

Design is a profession of thinkers and visionaries, people who shape human experience of the future by learning from the past and the present. They're trained by exploration and practice to spot patterns, trends and possibilities in people's day to day lives and gain insights from them. For these insights to be objective, meaningful and most importantly, actionable enough to evolve into ideas that improve human lives as well as the environment, a multidisciplinary field like design offers itself like a framework of effective problem solving.

The B.Des. course at UPID Noida is a 4-year full-time course in which candidates are admitted after 10+2 level examination or its equivalent as per eligibility guidelines of the AICTE/ University.

The aim of the undergraduate course is to develop skills, knowledge and attitude among the young design aspirants to become creative thinkers and problem solvers with a comprehensive value system. The value system here not only means social, moral and ethical values but also valuing our environment and the ecosystem.

The B.Des. course has three main focus areas- Product Design, User Interface and Experience Design (UI/ UX) and Visual Communication. The exposure in these areas is reinforced further through our pool of Electives which helps students to develop a broad understanding of the domain one wants to pursue as a profession. The entire curriculum has been drafted to develop competencies required as a Designer in a gradual manner that spreads across the four years.

Creative Thinking & Problem solving are among the fundamental skills required across diverse domains. The students begin developing these skills from the very first year through courses like- Introduction to Design and Design thinking which are then further developed in each semester through theme-based Design projects from the second year till the final year.

Skill Development and Enhancement is made in a chronological way with basics of communication and presentation being imparted through courses like- Creative visualization Techniques, Design Drawing, Technical Communication, Rendering and Illustration, in the first year. This is later enhanced through modules of Communication Studies in Design, Model Making and Hand Tools in second year. And further, managerial and entrepreneurial skills are developed through courses in the last two years.

Domain and Technical Knowledge plays a critical role and empowers a designer to conceptualise and translate ideas into reality. The students are exposed to the basics of Design in first year through courses like- Elements of Design, Principles of Design, Material Studies in Design etc. The knowledge of CAD, manufacturing and mechanics is imparted through modules on Computer Aided Design and Manufacturing, Kinetic art and Automata Design etc. in the second year. The last two years help the student to deep-dive into advance learning in these areas through core as well as the elective courses that complement the focus areas, one of them being the learning of Universal & Sustainable design.

Exposure is imperative in bringing a comprehensive perspective to a designer. The curriculum exposes the students to diverse domains associated with the design field including- Indian Art & Craft, Architectural Studies, On-field/ site visits through Design Documentation modules, Humanities & Social Studies, Environmental Sciences etc. The Industry connect is also ensured through internships that a student has to pursue in semester breaks and a final design project or thesis in the final semester where students are encouraged to do Industry-sponsored projects. In addition, the institute organises talk/ lecture series, workshops, contests etc. for the students during the four year course. The end semester evaluation, also known as Jury, is done by experts from the industry to bring in the aspect of industry feedback, for majority of the courses.

With these considerations in mind, the curriculum and detailed syllabus of Bachelor's Degree Course in Design is offered in this document with an intention to prepare graduates who can successfully render their services to the society, industry and the environment for everyone's benefit and their personal growth through a profession that is challenging yet exciting, philosophical yet pragmatic and emotional yet technology-driven.

EVALUATION SCHEME

SEMESTER I

Sl. No.	Course code	Subject	Periods			Internals			End Semester		Total	Credit
			L	T	P	CT	TA	Total	TE	PE		
1	CDS 111	Introduction to Design	1	0	8	30	70	100	0	100	200	**3**
2	CDS 112	Elements of Design	1	0	4	30	70	100	0	100	200	**3**
3	CDS 113	Materials and Processes in Design -I	1	0	4	30	70	100	0	50	150	**3**
4	CDS 114	Communication Studies in Design- I	2	0	2	30	70	100	0	50	150	**3**
5	CDS 115	Creative Visualization Techniques	1	0	8	15	35	50	0	100	150	**3**
6	CDS 151	Design Drawing	0	0	6	15	35	50	0	100	150	**3**
7		MOOCs (For B.Des. Hons Degree)*										**0**
		Total	**6**	**0**	**32**						**1000**	**18**

SEMESTER II

Sl. No.	Course Code	Subject	Periods			Internals			End Semester		Total	Credit
			L	T	P	CT	TA	Total	TE	PE		
1	CDS 121	Design Thinking	2	0	4	30	70	100	0	100	200	4
2	CDS 122	Principles of Design	1	0	8	30	70	100	0	100	200	3
3	CDS 123	Indian Art and Craft	2	0	0	15	35	50	30	20	100	2
4	CDS 124	Physical Ergonomics	1	0	6	30	70	100	0	100	200	4
5	CDS 125	Form Studies	1	0	4	15	35	50	0	50	100	3
6	CDS 141	Technical Communication	2	0	2	15	35	50	50	0	100	3
7	CDS 161	Rendering & Illustration	0	0	4	15	35	50	0	50	100	2
8		MOOCs (For B.Des. Hons Degree)*										0
		Total	**9**	**0**	**28**						**1000**	**21**

SEMESTER III

Sl. No.	Course Code	Subject	Periods			Internals			End Semester		Total	Credit
			L	T	P	CT	TA	Total	TE	PE		
1	CDS 211	Architectural Studies in Design - I ,Space	1	0	4	30	70	100	0	50	150	**3**
2	CDS 212	Communication Studies in Design- II	3	0	0	30	70	100	50	00	150	**3**
3	CDS 213	Model Making and Hand Tools Workshop	1	0	6	30	70	100	0	50	150	**3**
4	CDS 214	Computer Aided Design and Manufacturing -I	1	0	8	30	70	100	0	50	150	**3**
5	RVE 301	Universal Human Values and Professional Ethics	3	0	0	15	35	50	50	0	100	**3**
6	CDS 271	Design Documentation – I	0	0	2	0	50	50	0	50	100	**2**
7	CDS 281	Design Project – I, Simple Product Design	0	0	8	30	70	100	0	100	200	**4**
8	CDS 231	Cyber Security	2	0	0							**0**
9		MOOCs (For B.Des. Hons Degree)*										**0**
		Total	**11**	**0**	**28**						**1000**	**21**

SEMESTER IV

Sl. No.	Course code	Subject	Periods			Internals			End Semester		Total	Credit
			L	T	P	CT	TA	Total	TE	PE		
1	CDS 221	Design Management - I	2	0	2	30	70	100	0	50	150	**3**
2	CDS 222	Kinetic Art and Automata Design	1	0	4	15	35	50	0	50	100	**2**
3	CDS 223	Computer Aided Design and Manufacturing -II	1	0	4	30	70	100	0	50	150	**3**
5	CDS 261	Nature and Form	0	0	8	30	70	100	0	100	200	**4**
6	CDS 262	Tinkering Studio	0	0	8	30	70	100	0	100	200	**4**
7	CDS 291	Design Project – II, Display & Control Design	0	0	8	30	70	100	0	100	200	**4**
8	CDS 241	Environmental Science	2	0	0							**0**
9		MOOCs (For B.Des. Hons Degree)*										**0**
		Total	**6**	**0**	**34**						**1000**	**20**

SEMESTER V

Sl. No.	Course code	Subject	Periods			Internals			End Semester		Total	Credit
			L	T	P	CT	TA	Total	TE	PE		
1	CDS 311	Creative Narration	2	0	0	15	35	50	0	50	100	**2**
2	CDS 312	Humanities and Social Studies	2	0	0	15	35	50	50	0	100	**2**
3	CDS 313	Design Management – II	2	0	0	15	35	50	0	50	100	**2**
4	CDS 314	Design Research methodology - I	2	0	0	15	35	50	50	0	100	**2**
5	CDS 315	Architectural Studies in Design – II, Connectivity and Mobility	1	0	2	15	35	50	0	50	100	**2**
6	CDS 331	Elective – I	1	0	2	15	35	50	0	50	100	**2**
7	CDS 332	Elective – II	1	0	2	15	35	50	0	50	100	**2**
8	CDS 351	Design Workshop	0	0	8	15	35	50	0	50	100	**2**
9	CDS 381	Design Project – III, User Interface Design	0	0	8	30	70	100	0	100	200	**4**
10	CDS 333	Constitution of India	2	0	0							**0**
11		MOOCs (For B.Des. Hons Degree)*										**0**
		Total	**11**	**0**	**28**						1000	**21**

SEMESTER VI

Sl. No.	Course code	Subject	Periods			Internals			End Semester		Total	Credit
			L	T	P	CT	TA	Total	TE	PE		
1	CDS 321	Product Branding and Identity	2	0	2	30	70	100	0	50	150	**3**
2	CDS 322	Design Research methodology - II	1	0	2	30	70	100	0	50	150	**2**
3	CDS 323	Materials and Processes in Design - II	1	0	4	30	70	100	0	50	150	**3**
4	CDS 324	Sustainable Design	2	0	2	30	70	100	0	50	150	**3**
5	CDS 341	Elective - III	1	0	2	15	35	50	0	50	100	**2**
6	CDS 342	Elective - IV	1	0	2	15	35	50	0	50	100	**2**
7	CDS 391	Design Project – IV, Technically Complex Product Design	0	0	16	30	70	100	0	100	200	**4**
8	CDS 343	Essence of Indian Traditional Knowledge	2	0	0							**0**
9		MOOCs (For B.Des. Hons Degree)*										**0**
		Total	**10**	**0**	**30**						**1000**	**19**

SEMESTER VII

Sl. No.	Course code	Subject	Periods			Internals			End Semester		Total	Credit
			L	T	P	CT	TA	Total	TE	PE		
1	CDS 411	Professional Practice in Design	2	0	0	15	35	50	0	50	100	**2**
2	CDS 412	Universal Design	2	0	8	30	70	100	0	100	200	**4**
3	CDS 413	Packaging Design	1	0	4	30	70	100	0	50	150	**3**
4	CDS 431	Elective - V	1	0	2	15	35	50	0	50	100	**2**
6	CDS 471	Industrial Training	0	0	0	0	0	0	0	150	150	**3**
7	CDS 472	Design Documentation-II	0	0	2	0	50	50	0	50	100	**2**
8	CDS 481	Design Project – V, Systems Thinking in Design	0	0	16	30	70	100	0	100	200	**4**
9		MOOCs (For B.Des. Hons Degree)*										**0**
		Total	**7**	**0**	**30**						**1000**	**20**

SEMESTER VIII

Sl. No.	Course code	Subject	Periods			Internals			End Semester		Total	Credit
			L	T	P	CT	TA	Total	TE	PE		
1	CDS 499	Design Degree Project	0	0	40	0	500	500	0	500	1000	**20**
2		MOOCs (For B.Des. Hons Degree)*										**0**
		Total	**0**	**0**	**40**						**1000**	**20**

L/T/P: Lecture/ Tutorial/ Practical

CT: Class Test
TA: Teacher Assessment

TE: Theory Examination
PE: Practical Examination

* List of MOOCs (NPTEL) Based Recommended Courses for first year B. Des. Students:

1. Developing Soft Skills and personality-Odd Semester-8 Weeks-3 Credits

2. Enhancing Soft Skills and personality-Even Semester-8 Weeks-3 Credits

* After successful completion of 160 credits, a student shall be eligible to get Under Graduate degree in Design. A student will be eligible to get Under Graduate degree with Honours only, if he/she completes additional university recommended courses only (Equivalent to 20 credits; NPTEL Courses of 4 Weeks, 8 Weeks and 12 Weeks shall be of 2, 3 and 4 Credits respectively) through MOOCs. For registration to MOOCs Courses, the students shall follow NPTEL Site http://nptel.ac.in/ as per the NPTEL policy and norms. The students can register for these courses through NPTEL directly as per the course offering in Odd/Even Semesters at NPTEL. These NPTEL courses (recommended by the University) may be cleared during the B. Des. degree program (not necessary one course in each semester). After successful completion of these MOOCs courses the students, shall, provide their successful completion NPTEL status/certificates to the University (COE) through their college of study only. The student shall be awarded Hons. Degree (on successful completion of MOOCS based 20 credit) only if he/she secures 7.50 or above CGPA and passed each subject of that Degree Programme in single attempt without any grace marks.

Bachelor of Design | 1st Year (Semester I)

1. CDS 111- Introduction to Design

Course Code & Name	**CDS 111 - Introduction to Design**
Course Outcome	**To introduce the notion of design as it evolved through time; Understanding creativity & its application; Develop the ability to identify problems and finding needs; Understand the Design process**

Unit 1: Orientation
History & evolution of design, landmark discoveries & inventions, Design & its relationship with environment (art, craft, culture, society & technology), Various domains in design and design professions

Unit 2: Creative Thinking
Logical Thinking vs. Creative Thinking, Exploring creativity through various mediums, Interpreting creativity through Navrasa.

Unit 3: Problem Identification
Methods & Techniques, Observation, identification and analysis of a problem

Unit 4: Design Process
Understanding the design process, Exposure to various design terminologies, Case studies on Design process, Overview of system design

Unit 5: Design- Beyond product design
Introduction to Experience Design, Service Design, System Design etc.

References:
Don Norman; Design of Everyday Things, 2014
Don Norman; Emotional Design, Why we love (or hate) Everyday things, 2003
David Raizman; History of Modern Design, Prentice Hall,2004
William Lidwell, Kritina Holden, Jill Butler; Universal Principles of Design, 2003
Edward de bono; Lateral Thinking, 2010

2. CDS 112- Elements of Design

Course Code & Name	CDS 112 – Elements of Design
Course Outcome	Develop ability to explore, discover and understand the fundamentals used in design–its elements, features and principles

Unit 1: Line, Dot, Shape, Texture, Space, Form, Unity/harmony, Color

Its types and characteristics, characteristics of a shape, concepts of positive and negative space, types of shapes, visual and emotional interpretations

Unit 2: Coloring Primer

Primary colors and pure hues, Introduction to colors, pigment and light, additive and subtractive models. Shades of greys. Understanding warm and cold greys, Emotions and Colors Perception of colors, emotion and colors; pantone colors, colour wheel

Unit 3: Pattern recognition and creation

Pattern recognition, abstraction and construction. Developing patterns by repetition of points, dots, rectilinear elements, curvilinear elements, shapes, Regular and Irregular patterns. Fractals, Tessellation

Unit 4: Gradation and texturing

Understanding of flat surfaces. Material and process based textures, construction based textures, pigmented textures. Gradation on flat Surface

Unit 5: Creative art development with elements of design

Representation of elements in visual communication.

References:

Samara Timothy, Design Elements, 2nd Edition: Understanding the rules and knowing when to break them, Rockport Publishers, 2014
Evans Poppy and Thomas Mark A., Exploring the Elements of Design, Delmar Cengage Learning, 2012
Beech R., Origami – The Complete Guide to the Art of Paper Folding, Lorenz Books, 2001
Wong W., Principles of Two Dimensional Design, John Wiley & Sons, 1972
White Alex W., The Elements of Graphic Design, Allworth Press, 2011
Gail Greet Hannah, Elements of Design, Princeton Architectural Press, 2002

3. CDS 113- Materials & Processes in Design- I

Course Code & Name	CDS 113 - Materials & Processes in Design- I
Course Outcome	Knowledge of where most materials come from, Ability to identify which processes are used to make a product

Unit 1: Introduction to Materials
Classification of materials, History of significant materials.

Unit 2: Properties and Applications of materials
Properties and Applications Wood, bamboo, cane, leather ,fabric , jute, Steel, Brass, Bronze, Copper, Aluminium, Nickel, Tin, Lead, Zinc and Alloys etc.

Unit 3: Exploration of Materials
Exploration of materials such as fabrics, leather, wood and metals, Steel, Brass, Bronze, Copper, Aluminium, Nickel, Tin, Lead, Zinc and Alloys

Unit 4: Conventional manufacturing Processes
Manufacturing process introduction - Casting , Forging ,lathe, drilling ,milling ,welding, grinding, knurling, Foundry Tools and Equipments , Metal Cutting , other industrial practices

Unit 5: Basic hands on practices
Primary Importance of hands-on practices in product design; working with wood, paper, fabric, leather thread, wire, Acrylic sheets, sun board, fiber board, Introduction to making material boards.

References:
William D. Callister Jr., Materials Science and Engineering, Wiley, 2015
S. K. Hajra Choudhary and A. K. Hajra Choudhary, Elements of Workshop Technology Vol. I, MPP, 2000
C. Baillie and L. Vanasupa, Navigating the Materials World, Academic Press, San Diego, CA, 2003

4. CDS 114 - Communication Studies in Design- I

Course Code & Name	CDS 114 – Communication Studies in Design- I
Course Outcome	Ability to gain actionable insights from day to day conversations. Conceptual, and technical skills that make for effective communication design.

Unit 1: Understanding role of communication in culture
Defining and investigating communication, Social communication, Intercultural communication, Technology and culture, Effect of culture in communication, Models of communication, Ethics

Unit 2: **Communication Theories and Application**
Communication accommodation theory, Symbolic convergence theory, Relational dialectics, Agenda setting theory, Uncertainty reduction theory, Speech codes theory, Expectancy violation theory, Rhetoric, Cultivation theory etc.

Unit 3: **Storytelling in communication**
Speech codes, Interviews, Folk stories, Advertisements, TV shows and films, creating narratives etc.

Unit 4: **Practicing different mediums of communication**
Basics of Photography, Video, verbal and written language, gestures, audio, visual art etc.

Unit 5: Final project
Taking a topic and developing communication ideas for a community.

References:
Griffin, E. A first look at communication theory (8 th ed.). New York: McGraw-Hill, 2012
Lewis, J., Cultural studies: The basics. London: SAGE Publications, 2002
Watson, J, What is communication studies? London: Edward Arnold, 1985
Berko Roy, Basically Communicating .Wm. C. Brown Publishers, 1989
Roloff, M. E., & Miller, G. R, Interpersonal processes: New Directions in communication research. Newbury Park, Calif: Sage Publications, 1987
Carey, J. W, Communication as culture: Essays on media and society. Boston: Unwin Hyman, 1989

5. CDS 115- Creative Visualization Techniques

Course Code & Name	CDS 115 – Creative Visualization Techniques
Course Outcome	Ability to create compelling and detailed line drawings of real and imaginary objects; Develop skills of drawing and rendering from memory and imagination

Unit 1: Warm Up Exercises and Rapid Sketching
Representing the observed, representing concepts - Sketching for ideation; Lines; Geometric Shapes; introduction to colour and texture

Unit 2: Drawing Techniques
Grid based drawing, analytical representation; Inside-out sketching; Construction Drawing

Unit 3: Rendering and Sciography
Studies in light and shadow of 3-dimensional form representations; pencil rendering, Representing reality Mimetic Imagery and Abstraction; Sciography

Unit 4: Representing Imagination
Memory and Imagination; Object representation; Nature and life Representing nature; Figure drawing gestures and movements

Unit 5: Perspective Projection
One point, two point and three-point Perspective

References:
Betty Edwards, New Drawing on the Right Side of the Brain, 2002
Dalley Terence ed.,The complete guide to illustration & design, Phaidon, Oxford, 1980
T. C. Wang, Pencil Sketching, John Wiley & Sons,1997
Willy Pogany, The Art of Drawing, Madison Books, 1996
R. Kasprin ,Design Media – Techniques for watercolour, pen and ink, pastel and coloured markers, John Wiley & Sons,1999

6. CDS 151- Design Drawing

Course Code & Name	CDS 151 – Design Drawing
Course Outcome	**To familiarize students with drawing tools and accessories and to give them basic knowledge of good drafting and lettering techniques; To develop comprehension and visualization of geometrical forms. To familiarize with the concept of enlarging and reducing scales.**

Unit 1: Introduction
Importance of design drawing, Conventions and standards: ISO; Scales; Line types; Line Weights; Hatching Types; Lettering; Introduction to AutoCad

Unit 2: Projections of Points, Lines, Planes
Projection, Orthographic projections: Points, Lines, Planes (on paper & on Autocad)

Unit 3: Projection of Solids
Orthographic projections: Solids (on paper & on Autocad)

Unit 4: Sectioning & Intersection
Section of Solids, Intersection of Solids (on paper & on Autocad)

Unit 5: Axonometric views & Development of surfaces
Isometric view, Development of surfaces (on paper & on Autocad)

References:
A.J. Dhananjay, Engineering Drawing, TMH, 2008
N D Bhatt and V M Panchal, Engineering Drawing, 43rd Ed., Charotar Publishing House,2001
M B Shah and B C Rana, Engineering Drawing, 2nd Ed., Pearson Education, 2009
T E French, C J Vierck and R J Foster, Graphic Science and Design, 4th Ed., McGraw Hill, 1984.
W J Luzadder and J M Duff, Fundamentals of Engineering Drawing, 11th Ed., PHI, 1995
K Venugopal, Engineering Drawing and Graphics, 3rd Ed., New Age International, 1998

Bachelor of Design | 1st Year (Semester II)

1. CDS 121 - Design Thinking

Course Code & Name	**CDS 121– Design Thinking**
Course Outcome	**To develop the ability to comprehend problems in a creative way and come with solutions in a relatively shorter time frame.**

Unit 1: Design thinking process
Understanding Design thinking process and its stages, Various types of design approaches (models like double diamond, Stanford, IDEO, DeepDive, Etc), Convergent and divergent

Unit 2: Tools and techniques (Convergent)
3 "I"s (Influence, importance, imagination), ALU (advantages, limitations, unique potential), evaluation matrix etc

Unit 3: Tools and techniques (Divergent)
Brainstorming, Mind mapping, storyboarding, Point (pluses, opportunities, issues & new thinking), empathy map, group doodle, forced connections etc

Unit 4: Case studies for implementation and application
Product design, Service design, System design etc

References:
John Thackara, In the Bubble: Designing in a Complex World, The MIT Press, 2005
Bruce Hanington, Bella Martin, Universal Methods of Design: 100 Ways to Research
Vijay Kumar, 101 design methods, John Willey & sons, inc., 2013
Edward de bono, Lateral thinking, Penguin Books
Complex Problems, Develop Innovative Ideas, and Design Effective Solutions, Rockport Publishers, 2012
Donald A. Norman, Living with Complexity, MIT Press, 2010
Jeffrey Whitten and Lonnie Bentley, Systems Analysis and Design Methods, McGraw-Hill/Irwin, 2005
Gerald M. Weinberg and Daniela Weinberg, General Principles of Systems Design, Dorset House, 1988

2. CDS 122- Principles of Design

Course Code & Name	CDS 122– Principles of Design
Course Outcome	**Ability to create compelling colour schemes for graphic design** **Understanding the emotional impact of colours on people** **Ability to handle different types of pigments for model making and prototyping**

Unit 1: Principles of visual design
Principles of design, unity/harmony, balance, alignment, hierarchy; emphasis; similarity and contrast; Gestalt laws

Unit 2: Morphology
Scale, proportions; Movement, repetition; Pattern, rhythm, variety

Unit 3: Colour Psychology
Theory of Perception of colours; Emotion, colours and psychology; Color Psychology - Study of Hues as a determinant of human behavior.

Unit 4: Grays
Understanding the Grayscale, gradation methods; Composition with grays, black and white

Unit 5: Theory of Colour mixing
The Science of Colour Theories (Light & Pigment Theories); Colour Wheel; Tints, Tones and Shades, colour charts; Colour mixing models, colour palettes

References:
Itten J., The art of colour: the subjective experience and objective rationale of colour, John Wiley and Sons., 1974
Sherin, A., Design Elements, Color Fundamentals: A Graphic Style Manual for Understanding How Color Impacts Design, Beverly, Mass: Rockport Publishers, 2011
L. Hotzschue, Understanding Colour, VNR, 1995
R.M. Proctor, The principles of pattern, DoverPublications,1990
Elam, Kimberly; Geometry of Design: Studies in Proportion and Composition, Princeton Architectural Press, 2001

3. CDS 123- Indian Art and Craft

Course Code & Name	CDS 123- Indian Art and Craft
Course Outcome	Understanding of the history of art and culture of India through crafts and paintings. Also, gaining knowledge about its evolution over the years.

Unit 1: Introduction to art and craft history in India
Art and Craft, and the Interpretation of India's Past

Unit 2: Indian art movements
Introduction to Gandhara School of Art, Mathura School of Art etc.

Unit 3: Art and Craft in different states and GI marks
Indian art and craft mapped through different states and GI marks
(like Bidiri, Wood carving, Dhokra, Krishna shilla, Enamelling etc)

Unit 4: Indian Painting
An introduction to Indian paintings (Deccan painting, miniature paintings, etc)
Modern and contemporary painters

Unit 5: Changing Perspectives
Changing perspectives with globalization,
Field visits and report making

References:
Anil Rao, Sandhya Ketkar, The History of Indian Art, 2017
M. P. Ranjan, Aditi Ranjan , Handmade In India, 2007
10000 Years of Art, Phaidon Press, 2009
Dr. Alka Pande, Indian Art – The New International Sensation- A Collector's Handbook, 2008
Ilay Cooper and John Gillow, Arts and Crafts of India, 1996

4. CDS 124- Physical Ergonomics

Course Code & Name	CDS 124- Physical Ergonomics
Course Outcome	To introduce ergonomics and its significance in design. To understand the impact of anatomy, anthropometry, biomechanics, physiology and the physical environment on physical activity.

Unit 1: Introduction to Ergonomics
Definitions of Ergonomics and its classification, application and overview, Concept of Man Machine Environment System

Unit 2: Human body & its system
Overview of the Human Body and it's subsystems, Understanding musculoskeletal system and its function in terms of manual activities, Understanding nervous system, human sensory organs and their limitations.

Unit 3: Anthropometry
Understanding and applications of anthropometry; Basic anatomy, measurement system, types; static, dynamic, posture, joint, movement; Study of work posture and its impact on human performance; Physical environment and their impact on human performance

Unit 4: Product Ergonomics
Understanding of product ergonomics; Man, machine and Interaction

Unit 5: Safety
Injury prevention, safety, vibration, shock, fatigue and occupational hazard; Error handling

References:

Dr. Debkumar Chakraborty, Indian Anthropometric Dimensions For Ergonomic Design Practice, National Institute of Design,
R. S. Bridger, Introduction to Ergonomics, 2nd Edition, Taylor & Francis, 2003
J. Dul, and B. Weerdmeester, Ergonomics for beginners, a quick reference guide, Taylor & Francis, 1993
C. D. Wicknes, S. E. Gordon, and Y. Liu, An Introduction to Human Factors Engineering, Longman, New York, 1997
E. Grandjean, Fitting the task to the man, Taylor & Francis Ltd.1980
P.W. Jordan and W.S.Green, Human Factors in Product Design: current practice and future trends, Taylor & Francis, London, 1999

5. CDS 125 - Form Studies

Course Code & Name	CDS 125 – Form Studies
Course Outcome	**To understand form and its transformation; Develop ability to manipulate form for demonstration of varied expressions; Understand and develop family of forms with a common design language.**

Unit 1: Understanding of form
Definition of form, evolution of a flat shape into a volume, Classification of form 2D & 3D, Solids (Platonic, Archimedean)

Unit 2: Volume relationships
Dominant, subdominant & subordinate

Unit 3: Transformation
Radii manipulation, Form transition (addition & subtraction)

Unit 4: Form, Emotions & Identity
Abstraction & expression of form, Identity experimentations with form, texture & colour, Family of forms

Unit 5: Form explorations
Through different materials (Like- Paper Mache, thread, Plaster of Paris, Clay etc.)

References:
Gyorgy Kepes, Language of Vision, Dover Publications, 1995
Kimberly Elam, Geometry of Design: Studies in Proportion and Composition, Princeton Architectural Press, 2001
Gaston Bachelard and Maria Jolas (Translator),The Poetics of Space, Beacon Press; Reprint edition, 1994
Gail Greet Hannah, Elements of Design, Princeton Architectural Press, 2002
H. G. Greet and R. R. Kostellow, Elements of Design and the Structure of Visual Relationships, Architectural Press, NY, 2002

6. CDS 141 -Technical Communication

Course Code & Name	CDS 141 – Technical Communication
As per University curriculum	

7. CDS 161- Rendering & Illustration

Course Code & Name	CDS161 – Rendering & Illustration
Course Outcome	Ability to visualize ideas; Ability to do visual design explorations; To be able to project ideas in compelling manner from imagination to media

Unit 1: Introduction to Rendering
Types of drawing & rendering; Importance of rendering

Unit 2: Lights and its effects
Interaction with Light Highlights, shadow and reflection study of objects; Direct and indirect illumination

Unit 3: Photorealistic Visualization
Photorealistic Visualization, Rendering objects by observation, Rapid sketching techniques; Visual compositions of objects

Unit 3: Digital techniques in Rendering and Illustration
Digital sketching; Vector illustrations; RasterIllustrations; Introduction to Image processing softwares like Adobe, Corel Draw, Inkscape, GIMP; Digital Illustration Techniques Exposure and demonstration of Illustration and rendering software

Unit 4: Illustrations
Importance of text in illustrations; Story telling by illustrations; Illustrated drawings

References:
Rober McKim, Experiences in Visual Thinking, Brooks/Cole Publishing Company, 1980
Stephen Missal, Exploring Drawing for Animation (Design Exploration Series), Thomson Delmar Learning, 2003
D. K. Francis Ching, Design Drawing, John Wiley & Sons,1998
Tom Porter, Design Drawing techniques for architects, graphic designers and artists, Oxford Architectural Press,1991
Terence ed .Dalley, The complete guide to illustration & design, Phaidon, Oxford, 1980
T. C. Wang, Pencil Sketching, John Wiley & Sons,1997

Bachelor of Design | 2nd Year (Semester III)

1. CDS 211- Architectural Studies in Design – I, Space

Course Code & Name	CDS 211 –Architectural Studies in Design – I, Space
Course Outcome	Understand the learnings from history of art and design and demonstrate in the present day context; Understand the concept of space and its relationship with human behaviour.

Unit 1: Understanding Space
Typology of Spaces, Concept of Visual and Spatial continuity & Space hierarchy, Translation from 2D drawing/ illustration to 3D space

Unit 2: Art, Architecture & Design
Relationship in Architecture and Design, Why history is important? How design evolves? Different Art & Design movements

Unit 3: Space and human behaviour
Environmental psychology, Elements of spatial quality (Ex- Colour, pattern, texture, openings etc.), Various effects in space (Ex- Cathedral Effect, Biophilia effect, Proxemics etc.)

Unit 4: Space Design overview
Basics of - Interior Design, Exhibition Design, Scenic / Stage Design etc., Case Studies

Unit 5: Space Study
Documentation of space using framework
(Ex- AEIOU framework: Activities, Environments, Interactions, Objects, Users)

References:
William Lidwell, Kritina Holden, Jill Butler; Universal Principles of Design, 2003
Bruce Hanington, Bella Martin; Universal Methods of Design, 2012
Francis D.K. Ching, Architecture, Form, Space & Order, Fourth Edition
Yatin Pandya; Elements of Spacemaking, 2013
D. A. K. Kopec; Environmental Psychology for Design, 2006

2. CDS 212- Communication Studies in Design- II

Course Code & Name	CDS 212 – Communication studies in Design- II
Course Outcome	Understanding communication channels, Understanding semiotics and its cultural aspects.

Unit 1: Information Processing
Coding & Decoding; Sender, Channel and Receiver; Drawing as method of interaction, observation and documentation

Unit 2: Indirect communication
Study of relationships between Signifier, Signified and context; Denotation and Connotation; Communicating through gestures, voice, type and visuals

Unit 3: Introduction to Semiotic Perspective
Visual perception & semiotics; Sign: Concept and Types; Codes: Concepts, Types and Sharing; Usage of visual semiotics

Unit 4: Semiotic Interpretations and Cultural Metaphors
Myths- Concepts and Debates; Communication as Text / Discourse; Ideology: Link to Meaning Making; Calendrical events: understanding of festivals and rituals; Signs and their meanings in Indian cultures

Unit 5: Introduction to Rhetoric Perspective Origin and Evolution
Functions of Rhetoric; Key Elements of Rhetoric; Elements and analyzation of Rhetorical Presentation

References:
Ronald H. Forgus, Perception, The basic process in cognitive development, USA, McGraw-Hill 1996
Arthaya, Seminar on Visual semantics, IDC, IIT Bombay 1992
Carey, J. W. (1989). Communication as culture: Essays on media and society. Boston
Ghanekar, A (1998) Communication skill for effective management. Everest Publishing House. Gilligan,Pune
Fiske, J. (1982). Introduction to communication studies. London, Angleterre: Methuen
Schlenker, B. R. (1980). Impression management: The self-concept, social identity, and interpersonal relations. Monterey, Calif: Brooks/Cole Pub. Co

3. CDS 213 – Model Making and Hand Tools Workshop

Course Code & Name	CDS 213- Model Making and Hand Tools Workshop
Course Outcome	Understanding and working under workshop conditions; Understanding and specific tools for different jobs; Using conventional methods for product prototyping

Unit 1: Workshop tools & Practices
Introduction to workshop practices, Tools; metrology; machines and workshop work ethos; importance of PPE, safety procedures; and first aid

Unit 2: Working with plastics & Elastomers
Working with thermoplastics and thermosets; limitations; environmental impact; advantages and disadvantages of working with plastics.

Unit 3: Working with Ceramics and glass
Ceramics Types and Properties, Processing and use of ceramics and glass.

Unit 4: Working with Wood
Woodworking joints, Carving, and finishing

Unit 5: Working with Metal
Working with flat products and long products of steel, aluminium copper etc sizes and availability, joining techniques

References:
J. Garratt, Design and Technology, Cambridge University Press, UK, 20004 R. Thompson, Manufacturing processes for design professionals, Thames & Hudson, London 2007 Michael Ashby and Kara Johnson, Materials and Design: The Art and Science of Material Selection in Product Design, Butterworth Heinemann, 2002

4. CDS 214- Computer Aided Design and Manufacturing- I

Course Code & Name	CDS 214- Computer Aided Design and Manufacturing- I
Course Outcome	Ability to create Engineering CAD ready Surface and Solid Models, Ability to conduct design iterations in CAD software

Unit 1: **Computer as an Aid to the Designer**
Role of Computer in Designing; Computer Graphics, Systems and Hardware; Graphics Standards; Different types of Geometric Modeling Software

Unit 2: Modeling and Generation of Curves and Surfaces
Geometric figures and their representation; Types of curves and surfaces; Scanning and tracing Sketches; Modeling of curves and surfaces using software; Freeform surface modelling; Generating 2D designs using computers (engraving and routing)

Unit 3: Solid Modeling
Representation of solids: wireframe, B-rep and CSG; Modeling of simple solids using software; Modeling of complex solids using software; Generation of 2D drawings from 3D models

Unit 4: 3D Modeling of Assemblies
Modeling of Machine elements; Modeling of assemblies; Modeling of moving systems; Animation

Unit 5: Computer Aided Manufacturing and Project Work
Introduction to Computer Aided Manufacturing; Project Work in Modeling of a Product

References:
Ibrahim Zeid (2009), Mastering CAD/CAM, 2nd Edition, Tata McGraw Hill International Edition
P N Rao (2010), CAD/CAM Principles and Applications, 3rd Edition, Tata McGraw-Hill Education
Christoph M. Hoffmann, Geometric and Solid Modelling: An Introduction
William Howard and Joseph Musto. Introduction to Solid Modeling Using solid works. McGraw Hills
Alejandro Reyes, Beginners Guide to SolidWorks, SDC Publications

5. CDS 271- Design Documentation - I

Course Code & Name	CDS 271- Design Documentation - I
Course outcome	**Comprehensive understanding of the history and present state of one cottage industry/craft of Uttar Pradesh or India; Aptitude to conduct thorough field research and collect qualitative and quantitative information which may be useful for other industries and people**

Field study of Micro, Small Scale Industry/ NGO

- Field study of selected Micro, Small Scale industry /NGO to understand the functioning.
- Observe the complete process and figure out areas that can be improved upon from a designer point of view.
- Work on identified areas to make it better.
- Product design and development
- Documentation, Presentation of above
- Submission of Report

6. CDS 281- Design Project – I, Simple Product Design

Course Code & Name	BDS 281 – Design Project – I, Simple Product Design
Course outcome	To understand the process of design and be able to find solutions to simple problems by modifying forms and functions.

Unit 1: Overview of Design Process and its ecosystem
Design Relevance: Exposure and analysis, Case studies

Unit 2: Ideation
Brainstorming; Differential Discussion; group methods to generate ideas; solitary methods to generate ideas; Lateral Thinking

Unit 3: Concept Detailing
User Journey maps, User stories, activity mapping

Unit 4: Design Project
Design and Development of Product as per the brief

References:
D.Norman;The Design of Everyday things, London, The MIT Press, 1998
A.Forty; Objects of Desire, Thames & Hudson, 1995
J. de Noblet ed., Industrial Design- Reflections of a century,Thames & Hudson, 1993
Julier, G.;20th Century, Design,Thames & Hudson, 1993
Potter, Norman; What is a Designer: Things, Places, Messages, Princeton Architectural Press, 2002

7. RVE 301- Universal Human Values

Course Code & Name	RVE 301- Universal Human Values & Professional Ethics
As per University Curriculum	

8. CDS 231- Cyber Security

Course Code & Name	CDS 231 –Cyber Security
As per University Curriculum	

Bachelor of Design | 2nd Year (Semester IV)

1. CDS 221–Design Management - I

Course Code & Name	**CDS 221–Design Management- I**
Course Outcome	**Understanding of why people buy things, Ability to offer relevant products which people need, Awareness about Design Rights and their enforcement by law**

Unit 1: Design Management (scope for research)
Introduction of DM, Design Manager and its roles, objectives of DM, Pillars of Design management, Strategy & supply chain techniques, Roles of design in supply chain, impact of design on business performance, Scope for Research and Development in Design Management

Unit 2: An introduction to Consumer psychology
Definition and understanding of consumer, consumer needs, Identification of problems and user needs and Driving Factors; Emotional Design, Analysis of an existing problem in a given context, Social Ethics and Concerns

Unit 3: Market Study
Definition of market, consumer, buyer; Consumer Vs Buyer, Consumer Groups, Buyer Groups, Market Trends, Market Gaps

Unit 4: Business Communication
Explaining Ideas through briefs, detailed briefs and concept notes, Informal and Formal Business Communication

Unit 5: Project management
Managing projects, project organisation, project factor, project breakdown structure, project planning and phasing, project risk, project management tools

References:
Kathryn Best, The Fundamentals of Design Management, AVA Publishing, 2010
Brigitte Borja De Mozota, Design Management: Using Design to Build Brand Value and Corporate Innovation, Allworth Press, 2004
Kenneth B Khan, Product Planning Essentials, M E Sharpe Inc, 2011
John Stark, Product Lifecycle Management: 21st Century Paradigm for Product Realisation, Springer, 2011
Craig M. Vogel and Jonathan Cagan, Creating Breakthrough Products: Innovation from Product Planning Program Approval, FT Press, 2001
David L. Rainey, Product Innovation: Leading Change through Integrated Product Development, Cambridge University Press, 2011

2. CDS 222- Kinetic Art and Automata Design

Course Code & Name	CDS 222- Kinetic Art and Automata Design
Course Outcome	Taking inspiration from art, nature, human or animal and developing mechanical designs

Unit 1: History of Kinetic Art and Automata
History and development of kinetic Art & automata, Art Tradition to Modern, Interactive Art and Artistic, Optical Art, Visual Perception, Creation Bicycle Wheel, Kinetic Construction (Standing Wave) (1920) Arc of Petals (1941), Moving folk toys, etc.

Unit 2: Kinematics of Particles and Rigid Bodies
Kinematics of particles and rigid bodies; System of particles, Translation; Fixed; Axis rotational; General plane motion; rectilinear motion; curvilinear motion; Relative and constrained motion; Space curvilinear motion; etc.

Unit 3: Kinetics of Particles and Rigid Bodies
Kinetics of Particles and Rigid Bodies; Force, Work and Energy; Impulse and Momentum; Impact problems; Work energy; Power; Impulse-momentum and Associated conservation principles

Unit 4: Practical Mechanics
Exploring mechanical properties of model making materials through experiments; Introduction to DIY culture; Building mechanisms using workable materials; Exploring degrees of freedom, joints and links through experiments. 4 Bar Mechanisms, Cams, Gears and followers, levers and Linkages, Cranks, wheels, springs, pulleys, ratchets and pawls mechanism

Unit 5: Bio Mechanics
Skeletal mechanisms of Animal limbs, skeletal mechanisms of human limbs, structural characteristics of Plants, structural characteristics of exoskeletons

References:
Robert Race, Making simple Automata
Rodney Frost, Creative kinetics: Making mechanical marvels in wood
I.H. Shames, Engineering Mechanics: Statics and Dynamics, 4th Ed., PHI, 2002
F. P. Beer and E. R. Johnston, Vector Mechanics for Engineers, Vol I - Statics, Vol II – Dynamics, 3rd Ed., Tata McGraw Hill, 2000
J. L. Meriam and L. G. Kraige, Engineering Mechanics, Vol I – Statics, Vol II – Dynamics, 5th Ed., John Wiley, 2002
R. C. Hibbler, Engineering Mechanics, Vols. I and II, Pearson Press, 2002
S. Timoshenko and D. H. Young, Engineering Mechanics, McGraw Hill, 1956
J. Garratt, Design and Technology, Cambridge University Press, 1996

3. CDS 223-Computer Aided Design and Manufacturing-II

Course Code & Name	CDS 223- Computer Aided Design and Manufacturing-II
Course Outcome	Ability to understand the importance of CAD/CAM principles in the Product development, to model complex geometries using point cloud data and print 3D objects

Unit 1: Computer Aided Manufacturing
Details of different types of Computer Aided Manufacturing Systems, Role of CAM in today's world, Recent Advances in CAM

Unit 2: Geometric Modeling Using Point Cloud
Point cloud data acquisition, Point cloud data Handling, Fitting curves and surfaces through point Clouds, Design modification of 3D models from point cloud data

Unit 3: Rapid Prototyping (or 3D Printing)
Introduction to Rapid Prototyping (or 3D Printing), Rapid Prototyping Data Formats, Rapid Tooling

Unit 4: Rapid Prototyping Systems
Liquid based RP systems, Solid based RP systems, Powder based RP systems

Unit 5: Product Design and Manufacturing
(using Reverse Engineering and Rapid Prototyping)
Project Work

References:
Ibrahim Zeid (2009), Mastering CAD/CAM, 2nd Edition, Tata McGraw Hill International Edition
P N Rao (2010), CAD/CAM Principles and Applications, 3rd Edition, Tata McGraw-Hill Education
James A. Rehg and Henry W. Kraebber (2004), Computer Integrated Manufacturing, 3rd Edition, Pearson Education
Mikell P. Groover and Emory W. Zimmers (2003), CAD/CAM: Computer Aided Design and Manufacturing, Prentice Hall Edition
Liou W. Liou, Frank W. Liou, (2007), Rapid Prototyping and Engineering applications: A Tool Box for Prototype Development, CRC Press.
Ali K. Kamrani, Emad Abouel Nasr, (2006), Rapid Prototyping: Theory and Practice, Springer.
Peter D. Hilton and Paul F. Jacobs, (2000), Rapid Tooling: Technologies and Industrial Applications, Marcel Dekker Publication
Jacobs, P. F., (1996), Stereolithography and other RP&M technologies, ASME.

4. CDS 261- Nature and Form

Course Code & Name	CDS 261– Nature and Form
Course Outcome	Understand and acknowledge the design in Nature; To develop an ability to take inspirations from nature and use it in designing.

Unit 1: Understanding Nature & its design
Evolution of form in nature; Pattern & symmetry; Fibonacci series & the golden ratio

Unit 2: Form & Metaphor
Understanding metaphors; Use of visual metaphor in product; Metamorphism

Unit 3: Nature inspired Design
Understanding Biomimicry through form, features and its application; Case studies

Unit 4: Mini project
Hands-on Project/Assignments

References:

Maggie Macnab, Design by Nature: Using Universal Forms and Principles in Design, New Riders, 2011
Rudolf Finsterwalder, Form Follows Nature: A History of Nature as Model for Design in Engineering, Architecture and Art, Springer Vienna Architecture, 2011
Alan Powers, Nature in Design: The Shapes, Colors and Forms that Have Inspired Visual Invention, Conran, 2002
Ellen Lupton, Jennifer Tobias, Alicia Imperiale, Grace Jeffers and Randi Mates, Skin: Surface, Substance, and Design, Princeton Architectural Press, 2002
Mario Livio, the Golden Ratio: The Story of PHI, the World's Most Astonishing Number, Broadway, 2003

5. CDS 262– Tinkering Studio

Course Code & Name	CDS 262– Tinkering Studio
Course Outcome	Inculcate the habit of making; Checking feasibility by proof of concept; Thinking in real world terms; Design solutions with creative thinking; Using automation tools for creativity

Unit 1: Programming.
Basic Programming Concepts in C; Programming with open hardware systems like -Arduino; Raspberry Pi etc., Processing IDE

Unit 2: Electronics
Introduction to electronics and components; Resistors, Capacitors, Inductors, Switches, Diodes, Transistors, Sensors and Actuators. Their Usage and applications. Integration of sensors with Programmable Logic Controllers like Arduino, Raspberry Pi

Unit 3: Control Systems
Open Loop & Closed Loop; Feedback Control Systems, PID Control Systems; Logical controls; On-Off Controls; Physical realization of the controllers: hydraulic, pneumatic and electronic controllers; Using programmable Logic controllers

Unit 4: Lateral thinking
Lateral thinking by creating puzzle/ game board's designs and working solutions using newer or available technology/ DIY kits. Creative thinking by solutions to an existing real world problem.

Unit 5: Tinkering Projects
Tinkering with Soft and easily cut and moldable Materials; Thinking innovatively within resources

References:
Sclater, N Chironis, N. P. (2001). Mechanisms and mechanical devices sourcebook (Vol. 3). New York: McGraw-Hill. Ratan, S. S. Theory of machines. S Chand. Edward de Bono, Lateral Thinking, Penguin UK, 2010 Garratt J., Design and Technology, Cambridge University Press,1996 Don Norman, The Design of Everyday Things, Basic Books, 2014 Martin Gardner; 100 Perceptual puzzles, 1995 Barnes & Noble Books Pearson Education,2000; Yashvant Kanetkar, Let Us C

6. CDS 291 –Design Project – II, Display & Control Design

Course Code & Name	CDS 291 – Design Project – II, Display & Control Design
Course Outcome	Examine and evaluate man-object relationship and its interface; Develop capability to analyse, comprehend the current display & control systems.

Unit 1: **Orientation**
Need of Display & Control systems, Role of human senses, Components of Display & Control systems (Ex- input channels, response mechanisms, gratification models, action – reward etc.)

Unit 2: Interaction Design
About Interaction Design, User Interface and User Experience, Usability Evaluation- Heuristics, Fitts Law, Hick's law; Prototyping Tools/ Techniques –Physical & Digital

Unit 3: Ergonomics Overview
Human Factors – Physical & Cognitive ergonomics; Visual hierarchy & structure of information; Display & Controls composition through – color, placement, orientation etc.

Unit 4: Design Project
As per Design brief

References:
Wilbert O. Galitz, The Essential Guide to User Interface Design: An Introduction to GUI Design,Wiley
Jeffrey Anshel, Visual Ergonomics Handbook, Taylor & Francis group, 2005
William Lidwell, Kritina Holden, Jill Butler; Universal Principles of Design, 2003
Bruce Hanington, Bella Martin; Universal Methods of Design, 2012
Dix, Alan J.; Finlay, Janet E.; Abowd, Gregory D.; Beale, Russell; Human-Computer Interaction, Pearson Education; 2 edition, 1998
Nielson, Jackob; Usability Engineering; Morgan Kaufmann, 1993

7. CDS 241 – Environmental Science

Course Code & Name	CDS 241 – Environmental Science
As per University Curriculum	

Bachelor of Design | 3rd Year (Semester V)

1. CDS 311- Creative Narration

Course Code & Name	**CDS 311– Creative Narration**
Course Outcome	**Ability to comprehend and draft user stories, Ability to document user experience in a retainable form.**

Unit 1: Story
Story, narrative and meaning making, metaphors

Unit 2: Objective
Premise and problem statement, Context

Unit 3: Protagonist
Characters and personas, examples of Don Quixote, Karna

Unit 4: Chain of Events
Plot and Scenarios

Unit 5: Dynamics
Relationship between problems, need and conflict, rationalization of need, rationalization of conflict, Action and Resolution

References:
Mike Korolenko and Bruce Wolcott, Storytelling and Design: Media Literacy for the Digital Age, Pearson Learning Solutions, 2005
Marie-Laure Ryan (editor), Narrative across Media: The Languages of Storytelling, University of Nebraska Press, 2004
Kristin M. Langellier and Eric E. Peterson, Storytelling in Daily Life: Performing Narrative, Temple University Press, 2004

2. CDS 312- Humanities And Social Studies

Course Code & Name	CDS 312– Humanities And Social Studies
Course Outcome	**Develop visual awareness of the present-day environment. Recognize and relate design forms to historical precedents and possible future developments, Have a greatly increased general knowledge, including source material which one can draw on for future student and development in the field of design**

Unit 1: Artistic Creation
Complete understanding of Perception, Communication, Imagination, Expression, and Creativity for artistic creation.

Unit 2: Expression through Art & Design movements
Understand the thoughts and techniques involved in important art movements- impressionism, cubism, constructivism, surrealism, romanticism etc. Students are expected to express their ideas via Posters, Murals, Building Art, Collage, Graffiti, 3D-Installations.

Unit 3: Anthropology
Defining anthropology, Branches of anthropology- Physical anthropology, Social Anthropology, Archaeological Anthropology, Environmental Anthropology, Anthropology and Methods of Research.

Unit 4: Ethnography
Observations/Analysis; Community- Based Ethnographic Research; Activity Theory; Empathy in Design; Value Sensitive Design; Historical development of fieldwork; relations between field methods and dominant theoretical orientations; varieties of fieldwork at present; the implications; Ethnographic research design as a continuous process; the formulation of research problems.

Unit 5: Introductory Sociology & Psychology
Sociology as a Science of Human Society: Introduction:- Basic concepts (Roles, Norms, Values, Groups and Institutions), Social Structure, Culture, Perspectives (Functionalist, Conflict & Interactionist), Psychological Perspectives and Approaches

References:
Geraldine Gay and Helene Hembrooke, 2004, Activity-Centered Design, An Ecological Approach to Designing Smart Tools and Usable Systems.
Amy. E. Aniston, Graphic Design Basics (IInd Edition)
Lydia Darbyshire, Practical Graphic Design Technique
Clifford Geertz, Interpretation of Cultures
Batya Friedman and Alan Borning, Value Sensitive Design and Information Systems
Julian Murchison, Ethnography Essentials: Designing, Conducting, and Presenting Your Research

3. CDS 313- Design Management–II

Course Code & Name	CDS 313– Design Management–II
Course Outcome	Ability to communicate novel ideas to general stakeholders

Unit 1: Product Lifecycle Management
Introduction to design management, Understanding & importance of PLM, Phases of PLM, Understanding of steps of product life cycle, procurement, Process monitoring, Quality Assurance, Guarantee Statement, Warranty Statement

Unit 2: Business Development
Understanding of Business, Forecasting, importance and needs of forecasting, process of forecasting, Product Planning for the future, Disruptive Innovation; Observations, insights and the opportunities

Unit 3: Marketable Design Development and its Accounting
Design development for the market, measuring value of design, types of budget, budget planning, cost calculation for pricing structure, measuring performance.

Unit 4: Activities in Business Management
Discussion on Knowledge Management, Induction, Scale & scope of Work, Work distribution, Time management, Scheduling, Feedback, Information loops, updates, record keeping documentation.

Unit 5: Intellectual Property Rights
Overview of Intellectual Property Rights and its importance, Patent, Copyright, Trademark, Trade secrets, Industrial Design, Geographical Identification, IPR Registration processes

References:
Best, K. (2006). Design management: managing design strategy, process and implementation. AVA publishing
Cooper, R., Junginger, S., & Lockwood, T. (Eds.). (2013). The handbook of design management. A&C Black
Kathryn Best, The Fundamentals of Design Management, AVA Publishing, 2010
Brigitte Borja De Mozota, Design Management: Using Design to Build Brand Value and Corporate Innovation, Allworth Press, 2004
John Stark, Product Lifecycle Management: 21st Century Paradigm for Product Realisation, Springer, 2011
Craig M. Vogel and Jonathan Cagan, Creating Breakthrough Products: Innovation from Product Planning Program Approval, FT Press, 2001
David L. Rainey, Product Innovation: Leading Change through Integrated Product Development, Cambridge University Press, 2011

4. CDS 314- Design Research Methodology –I

Course Code & Name	CDS 314– Design Research Methodology –I
Course Outcome	Ability to be able to understand the basics of research and techniques.

Unit 1: Why Do We Do Research?
Research: an integral part of your practice (basic introduction to research)

Unit 2: Basic Techniques
The Research Process: A Quick Glance: Deciding what to research, Planning a research study (sampling techniques and data collection), qualitative and quantitative techniques.

Unit 3: Literature Review
Reviewing the Literature

Unit 4: Problem Identification
Formulating a Research Problem by identifying scope for research

Unit 5: Report / Paper Writing (An Introduction)
Colloquium paper writing

References:
Brenda Laurel, Design Research: Methods and Perspectives, The MIT Press, US, 2003
R. D. Wimmer & J. R. Dominick, Mass media research: An introduction. Belmont, California, Wadsworth Pub. Co., 2000
A. Hansen, Mass communication research methods. New Delhi: Log Angeles, 2009
E. R. Babbie, The practice of social research. Belmont, California, Wadsworth Pub. Co., 1992
C. R. Kothari, Research methodology: Methods & techniques. New Delhi: New Age International (P) Ltd., 2004

5. CDS 315- Architectural Studies in Design–II, Connectivity and Mobility

Course Code & Name	CDS 315 –Architectural Studies in Design–II, Connectivity and Mobility
Course Outcome	To develop an appreciation & expose the students to the history and development of planning / concepts, its relevance & application to modern day principles of Spatial Planning.

Unit 1: Connectivity
Communication channels; interpersonal communication; mass communication; Public Information systems, Internet of Things, Artificial Intelligence.

Unit 2: Movement
Modes of Transportation; Individual Movement; Mass movement; Transportation as service; Transit Systems; Commuter centric Design.

Unit 3: Planning concepts & Infrastructure
Planning concepts related to City beautiful movement, Remote Area Connectivity; Rural Transportation; Rural Road networks, Urban Connectivity; High Density Transportation; Transit Oriented Development; Public Safety, Accessible Design; Universal Design in Public Transportation.

Unit 4: Roads and traffic studies
Awareness of concepts related to various traffic problems in India, Understanding of PCU, Traffic volume, Road capacities, Road types; their sections and intersections, Traffic calming as per IRC guidelines.

Unit 5: Sustainable and Futuristic Transportation
Structure and design flaws of Public Transportation, Alternative Energy resources for transportation; Introduction of Electric Vehicles.

References:
J. Dul, and B. Weerdmeester, Ergonomics for beginners, a quick reference guide, Taylor & Francis, 1993
C. D. Wicknes, S. E. Gordon, and Y. Liu, An Introduction to Human Factors Engineering, Longman, New York, 1997
E. Grandjean, Fitting the task to the man, Taylor & Francis Ltd.1980
Dr. Debkumar Chakraborty, Indian Anthropometric Dimensions For Ergonomic Design Practice, National Institute of Design, 1997
Arthur B. Gallion and Simon Eisner, The Urban Pattern – City planning and Design, Van Nostrand Reinhold company
N.V.Modak, V.N.Ambedkar, Town and country planning and Housing, orient longman, 1971
IRC Guidelines

5. CDS 351- Design Workshop

Course Code & Name	CDS 351 – Design Workshop
Course Outcome	Application of design process and developing forms and product ideations.

Unit 1: Inspiration and theme selection
Working on taking inspiration and theme. (Theme is the larger aspect, through brainstorming the idea, we can narrow down the aspect. Taking forward one idea as inspiration and working on studying in detail about the selected inspiration.)

Unit 2: Design boards and references
Brainstorm on all the ideas the student is working one, developing the supporting boards. (theme board, inspiration board, material board, client board, client profile, market study, mood boards, collecting as many reference images as possible etc.)

Unit 3: Form and concept generation
Developing forms through the reference images and boards. The forms have to be in the form of sketches, the students can apply concepts understood through the selected inspiration.

Unit 4: Concept exploration and development
Developing 3D Forms from the selected form sketches, using a variety of materials like plaster of Paris, wires, paper, cardboard, clay, foam, waste material etc.

Unit 5: Concept detailing in products and final project
Through the forms and sketches the students can design products incorporating concepts along with studying the market; Final display of product sheets with technical product dimensions, materials, textures, working and client details.

References:
Williams, R., & Sheldon, C. Robin Williams Handmade Design Workshop: Create Handmade Elements for Digital Design. Peachpit Press., 2009
Cuffaro, D., & Zaksenberg, I. The Industrial Design Reference & Specification Book: Everything Industrial Designers Need to Know Every Day. Rockport Publishers.2010
Sherwin, D.Creative workshop: 80 challenges to sharpen your design skills. How Books.2013
Fullerton, T., Swain, C., & Hoffman, S. Game design workshop: Designing, prototyping, & playtesting games. CRC Press. 2004
Vijay Kumar, 101 design methods, John Willey & sons, inc., 2013
Karl Aspelund, The design process, Fairchild Books, 2014
Alex Milton & Paul Rodgers, product design, Laurence King Publishing, 2011

6. CDS 381- Design Project- III, User Interface Design

Course Code & Name	CDS 381 – Design Project- III, User Interface Design
Course Outcome	Understanding the UX design; Development of proficiency in user experience in software interfaces; Understanding physiological and psychological considerations in UX design

Unit 1: Context, Content & User

Understanding Context, content & User (Contextual Enquiry, Interviews, etc.) for understanding the user and his requirements. Understanding the segmentation in and need of various user groups.

Unit 2: Physio-Psychology

Physio-psychological Constraints in designing; User Experiences Research Understanding the factors that define user experience.

Unit 3: Personas

Creating Personas; quantitative and qualitative study; Understanding the representative user; Understanding Personalities; Using models and tools to create personas; typical representative tasks

Unit 4: Mapping hierarchy

Understanding the needs-hierarchy of the user; Identification and Stratification of the users need in the interface; Information and control grouping and labeling; need to self-actualise

Unit 5: Design & Optimization

Design of visual interfaces, expressive interfaces, audio interfaces; Testing and optimizing the user interfaces; personality traits.

Design of interactive systems, products for future use, Collaborative products to be used in groups, devices for rural applications and devices for use in public places

References:

Donald A. Norman, Invisible Computer: Why Good Products Can Fail, the Personal Computer Is so Complex and Information Appliances Are the Solution, MIT Press, 1998 Brenda Laurel, Computer as Theater, Addison-Wesley Pub Co, 1993 Jef Raskin, The Humane Interface: New Directions for Designing Interactive Systems, Pearson Education,2000

7. CDS 333- Constitution of India

Course Code & Name	CDS 333 – Constitution of India
As per University curriculum	

ELECTIVES (Semester V)

Elective - I	CDS 331A- Technology for Digital Experience
	CDS 331B- Animation Design
	CDS 331C- Product Ergonomics & Styling
Elective - II	CDS 332A- UI Design and Development
	CDS 332B- Graphics and Typography
	CDS 332C- Furniture and Interior Design

Bachelor of Design | 3rd Year (Semester VI)

1. CDS 321- Product Branding and Identity

Course Code & Name	**CDS 321– Product Branding And Identity**
Course Outcome	**Understanding of why organizations invest in maintaining their identity, Knowledge of Branding and Identity design process**

Unit 1: Principles of Visual Design Refresher
Understanding visual culture; Visual Theories; Visual Design; Symbolism, Time, Sound; Point of View

Unit 2: Visual Art
Visual art History; Sculpture; Artistic Styles

Unit 3: Aesthetic Experience
Modes of Aesthetic Experience; Basics of Aesthetic values; Aesthetics of Thinking and Creativity; Taste and Aesthetes; Aesthetics of Symbols and Language;

Unit 4: Visual Experience
Photography and Moving Images; Historical, Technical and Cultural Perspective; Ethical and Critical Perspective; Motion Pictures; Television and Video; Reality Shows;

Unit 5: Branding and Identity
History of branding; structure of a Brand; Brand language; Logos; Copywriting; Typeface; Brand Placement; Brand Guidelines;Structure of Identity; Visual Abstraction; Metaphors; Communication; Representativeness; Evolution;

References:
Paul M., Visual Communication: Images with Messages, 2006
Ralph E. Wileman, Visual Communication
David Sless, Learning & Visual Communication
Friedrich O. Huck and Carl L. Fales, The Digital Evolution: Visual Communication in the electronic age
Zia-Ur-Rehman, Visual Communication an Information Theory Approach
Margaret Mark, Carol Pearson, the Hero and the Outlaw: Building Extraordinary Brands through the Power of Archetypes, McGraw Hill, 2001

2. CDS 322- Design Research Methodology –II

Course Code & Name	CDS 322– Design Research Methodology –II
Course Outcome	Ability to be able to write colloquium paper by applying the understanding of research in various areas.

- Applications of various techniques, report and colloquium paper writing.
- Informed opinion and judgment, market research,
- On various areas using Data Visualization techniques (for example Data Point, Bar, Stack, Pie, Donut, and Dot charts; Histograms) and thereby creating infographics reports on design.

References:

Brenda Laurel, Design Research: Methods and Perspectives, The MIT Press, US, 2003
R. D. Wimmer & J. R. Dominick, Mass media research: An introduction. Belmont, California, Wadsworth Pub. Co., 2000 A. Hansen, Mass communication research methods. New Delhi: Log Angeles, 2009 E. R. Babbie, The practice of social research. Belmont, California, Wadsworth Pub. Co., 1992 C. R. Kothari, Research methodology: Methods & techniques. New Delhi: New Age International (P) Ltd., 2004

3. CDS 323- Materials and Processes in Design- II

Course Code & Name	CDS 323– Material And Processes In Design- II
Course Outcome	Ability to design products which are seemingly impossible to manufacture using conventional processes

Unit 1: Introduction to Non-conventional Manufacturing Processes
Introduction to Non-conventional Manufacturing Processes and their classification, their role in design and industry

Unit 2: Additive and Material Deposition Processes
Laser Deposition, Micro-Plasma Powder Deposition, Chemical vapor Deposition, Micro Welding, Powder Casting, Metal 3D Printing, Powder Deposition 3D printing; Extruded Filament 3D printing, Clay 3D printing, Stereolithography

Unit 3: Subtractive and Cutting Processes
Electrochemical machining, Electro-Discharge machining, Ultrasonic Machining, Laser Beam Machining, Water jet machining, Abrasive Jet Machining, Plasma Arc machining, Water Jet Cutting, Plasma Cutting, Laser Cutting, Electro-Discharge Wire Cutting; Abrasive Jet Cutting

Unit 4: Special Purpose Manufacturing processes
Roto molding, Layer Compression, Sheet contouring, Friction Welding

Unit 5: Surface Treatment Processes
Laser Etching, Acid/Base Etching, Electrochemical Etching, Sandblast Etching, Ultraviolet Etching, Photochemical Machining, Electrochemical Polishing

References:
Vijay K. Jain, Advanced Machining Processes. Allied Publishers, New Delhi, 2007
P. C. Pandey and H.S. Shan, Modern Machining Processes, Tata McGraw-Hill, New Delhi, 2007
G.F. Benedict, Nontraditional Manufacturing Processes, Marcel Dekker Inc., New York, 1987
McGeough, Advanced Methods of Machining, Chapman and Hall, London, 1998
Paul De Garmo, J.T. Black, and Ronald A. Kohser, Material and Processes in Manufacturing, Prentice Hall India, 2001.

4. CDS 324- Sustainable Design

Course Code & Name	CDS 324- Sustainable Design
Course Outcome	To be able to understand factors that have an impact on the environment and sustainability in design.

Unit 1: Basics of Sustainable design
Introduction of sustainability, sustainable design principles. R's approach, Physical, mental, spiritual, cultural, social, ethical and economic issues in designing for sustainability, Planet centric Design, C2C approach

Unit 2: Impact of unsustainable practices on Environment
Ecological footprints, ecosystem impact.

Unit 3: Issues in design with respect to environment
Design and social concerns.

Unit 4: Sustainable Design & innovation
Frugal Design, Design for Environment/Eco design, Design for sustainability, Design from Waste, Eco innovation, system-wide product/service strategies

Unit 5: Lean Manufacturing
About Lean manufacturing and its principles, Tools & Techniques, Understanding through Success stories – Toyota, Nike, Intel etc.

References:
Mendler, S., & Odell, W. (2000). The HOK guidebook to sustainable design. John Wiley & Sons. 4.
Williams, D. E. (2007). Sustainable design: Ecology, architecture, and planning. John Wiley & Sons.
Walker, S. (2012). Sustainable by design: Explorations in theory and practice. Routledge.
Fairs, M. (2009). Green design: creative sustainable designs for the twenty-first century. North Atlantic Books.
James P. Womack, Daniel T. Jones (1996),Lean Thinking: Banish Waste and Create Wealth in Your Corporation
Jeffrey K. Liker, David Meier (2005), The Toyota Way Fieldbook, McGraw-Hill Education

5. CDS 391- Design Project- IV, Technically Complex Product Design

Course Code & Name	CDS 391- Design Project- IV, Technically Complex Product Design
Course Outcome	Ability to break down complex product ideas into smallest components, Ability to build functional prototypes

Unit 1: Orientation
Product Integration; Forward Integration; Backward integration; Top – Down Design Approach; Bottom-Up Design Approach; Reverse Engineering; Proof Of concept;

Unit 2: Development of creative concepts
With explorations of alternative solutions by mapping the functional requirements w.r.t. user requirements, ergonomics, functions, materials and processes.

Unit 3: Design Project
As per brief

References:
Steven Marjieh, Reverse Engineering Foundations: Product Design ,2018
Mike Baxter, Product Design, 1995
Ali K. Kamrani, Sa'Ed M. Salhieh, Ph.D., Sa'ed M. Salhieh, Product Design for Modularity, 2002

6. CDS 343- Essence of Indian Traditional Knowledge

Course Code & Name	CDS 343– Essence Of Indian Traditional Knowledge
As per AICTE Curriculum	

ELECTIVES (Semester VI)

Elective - III	CDS 341A- Gamification and UX
	CDS 341B- Photography and Image Processing
	CDS 341C- Toy and Games Design
Elective - IV	CDS 342A- Usability Testing
	CDS 342B- Film and Documentary
	CDS 342C- Mobility and Vehicle Design

Bachelor of Design | 4th Year (Semester VII)

1. CDS 411- Professional Practice In Design

Course Code & Name	**CDS411–Professional Practice In Design**
Course Outcome	**Expose students to professional practise in design; Looking at Design from a business perspective**

Unit 1: Entrepreneurship

Social Entrepreneurship, Business Entrepreneurship, Trading Entrepreneurship, Corporate Entrepreneurship, and Agricultural Entrepreneurship

Unit 2: Business Foundation

Timmons Model of Entrepreneurship, Investment Models, Startup Business Models, Business Plans, Pitch presentations, Small Business models

Unit 3: Legal aspects of business

Contracts and Agreements, Conflict Resolution, Arbitration

Unit 4: Running a Design Business

Set up of an independent design business, Hiring processes, Project Scheduling and work delegation, Cost Estimation; Billing, salaries and taxation

Unit 5: Professional Ethics

Ethics in Profession, Code of conduct

References:

Riadh Habash, Green Engineering: Innovation, Entrepreneurship and Design, 2017
Ted Crawford, AIGA Professional Practices in Graphic Design, Allworth Press, 2008
Douglas Davis, Creative Strategy and the Business of Design, 2016
Shan Preddy, How to Run a Successful Design Business: The New Professional Practice, Gower Publishing, Ltd., 2011
Min Basadur, Michael Goldsby, Design-Centered Entrepreneurship, 2016

2. CDS 412- Universal Design

Course Code & Name	CDS 412 – Universal Design
Course Outcome	Introduce the concept of Universal Design as a design approach; Exposure to accessibility standards and research techniques

Unit 1: **Orientation**
What is Universal Design? Its Origin and emergence, Need and its relevance today, Examples of UD.

Unit 2: Principles and Goals
Principles of Universal Design, Goals of Universal Design, Understanding above from various design spectrums- Product, building, information, service

Unit 3: Deep Dive
Universal - Inclusive - Accessible Design, Universal Design for Learning (UDL), Use of Assistive technologies, Research techniques like- AGNES (Age gain now empathy system simulation suit), Current ecosystem, Government Initiatives & policies.

Unit 4: Accessibility Standards and Guidelines
Physical accessibility standards for Barrier free environment, Web accessibility, WAI (Web Accessibility Initiative), World Wide Web Consortium (W3C)

Unit 5: Mini Project
As per brief

References:
Edward Steinfeld and Jordana L. Maisel, Universal Design – Creating Inclusive Environments, 2012
William Lidwell, Kritina Holden, Jill Butler; Universal Principles of Design, 2003
Bruce Hanington, Bella Martin; Universal Methods of Design, 2012
James Holmes-Seidle , Barrier-Free Design, 1996
CPWD, Guidelines and space standards for Barrier free built environment for Disabled and Elderly Persons, 1998

3. CDS 413- Packaging Design

Course Code & Name	**CDS 413– PACKAGING DESIGN**
Course Outcome	**Design packaging with consideration to various factors like branding, budget, target audience etc., Understanding connections between packaging, branding and the target audience.**

Unit 1: Introduction
Introduction to different kinds of packaging material; Studying about requirements of different products; Exposure of products, services and packaging in stores and similar environments

Unit 2: Case studies
Studying a variety of packaging case studies for different brands, materials used; studying the sustainability aspect; understanding the feasibility; Deconstructing and studying packaging of different brands (2-5 brands can be taken in consideration); Elaborating the study on any of the selected brands.

Unit 3: Project selection
Selection of product for packaging development; Researching about the current packaging available; constraints and positive aspects; understanding, target audience, budget, branding

Unit 4: Project development
Developing a packaging for the selected product; Incorporating improvements that can be worked upon after the research work.

Unit 5: Final packaging and its branding
Working on the branding aspect of the packaging that has been developed.

References:
Marianne R. Klimchuk, Sandra A. Krasovec, Packaging Essentials: 100 Design Principles for Creating Packages (Design Essentials), Rockport Publishers; 1 edition, June 1, 2010
Marianne R. Klimchuk, Sandra A. Krasovec, Packaging Design: Successful Product Branding from Concept to Shelf, John Wiley & Sons, 2006
Paul Jackson, Structural Packaging: Design your own Boxes and 3D Forms (Paper engineering for designers and students), Laurence King Publishing, 2012
Giles Calver, What Is Packaging Design? (Essential Design Handbooks), Rotovision, 2007
Peng Chong (Editor), Interactive Packaging Design, Design Media Publishing Ltd, 2018
Pentawards (Editor), The Package Design Book 2 (VARIA), TASCHEN; Mul edition, 2013
Tony Ibbotson, Peng Chong, Eco Packaging Now , Images Publishing Dist Ac, 2016

4. CDS 471- Industrial Training

Course Code & Name	CDS 471 – Industrial Training
Course Outcome	Attain hands-on experience on a live-project

Students would avail the opportunity to work with Design Firm/ Studios/ Industry under their guidance for a period of 6-8 weeks. The students are required to make a presentation of work done, submit a report and a letter/ certificate of their training.

5. CDS 472- Design Documentation- II

Course Code & Name	CDS 472- Design Documentation- II
Course Outcome	Exposure to site-specific study and their relevance; Documentation of Site visits and analysis, Ability to identify intervention area and come up with design solution

Unit 1: Introduction

Introduction to Site specific study; Studying about various sites like museums, historical sites, market places, villages etc.; understanding aspects to be covered and documentation details; compilation etc.

Unit 2: Case studies

Studying case studies for different site-specific projects already done; understanding various aspects; how the project started and its relevance etc.

Unit 3: Project

As per brief

Unit 4: Site Visit

The students will be visiting site as per brief conducting a study and document the same

Unit 5: Final presentation

Students are expected to present their site specific study and design solutions.

References:

Diana Panke, Research Design & Method Selection: Making Good Choices in the Social Sciences, 2018

Hemanta Doloi, Ray Green, Sally Donovan , Planning, Housing and Infrastructure for Smart Villages

6. CDS 481- Design Project – V, Systems Thinking in Design

Course Code & Name	CDS 481 – Design Project – V, Systems Thinking in Design
Course Outcome	**Understand how to design for applications that require a whole lot of variations and adaptability; Ability to come up with solutions that are sustainable, cross disciplinary and consider multiple perspectives.**

Unit 1: Orientation
Design thinking & System thinking, System thinking in Design, Understanding Systems and its typology

Unit 2: Theories and approaches
The fifth discipline approach, Role of Organizational Ergonomics etc.

Unit 3: Design Project
As per Design brief

References:
Peter M. Senge, The Fifth Discipline: The Art & Practice of the Learning Organization, 2006
John Thackara, In the bubble: Designing in a complex world, MIT Press, 2005
Donald. A. Norman, Living with complexity, MIT Press, 2010
Gerald M. Weinberg, Daniela Weinberg; General Principles of Systems Design, Dorset House, 1998
William Lidwell, Kritina Holden, Jill Butler; Universal Principles of Design, 2003
Bruce Hanington, Bella Martin; Universal Methods of Design, 2012

ELECTIVES (Semester VII)

Elective - V	CDS 431A- Omni-channel experience design
	CDS 431B- Image, Text and sound
	CDS 431C- Display and Space Design

Bachelor of Design | 4th Year (Semester VIII)

1. CDS 499- Design Degree Project

Course Code & Name	**CDS499–Design Degree Project**
Course Outcome	**Ability to apply and demonstrate all the skills and knowledge acquired in the previous semesters to solve a design problem**

Students in this semester would take sponsored /self-sponsored full-time Design projects. The project can be taken as per interest area of student or Industry project.

The project can be sponsored by industry or organisations and even self-sponsored.